

Where Clinicians Look Tells Us What to Measure: Attention-Defined Gait Measurements and Evidence-Gated Fusion for Adult Spinal Deformity

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Abstract. Clinician region annotations are usually injected into gait models as attention supervision. On lateral-view gait videos of adult spinal deformity (ASD) patients we show that this buys interpretability but not accuracy: a region-grounded 3D CNN reproduces clinician attention four times better than Grad-CAM while patient-level balanced accuracy stays at 0.64–0.66, a video transformer does no better, and four skeleton models are at chance. We instead let the annotations decide *what to measure*: the regions clinicians mark define five sagittal quantities from 3D keypoints, and a logistic regression on these five numbers reaches 0.72 (AUC 0.79; 67 patients, 10 random splits)—above every end-to-end model, above five random measurements (98th–100th percentile) and four quantities at unmarked regions (0.66), and equal to the best data-driven selection over 1,272 candidates. We then fuse the five numbers with the video model by an evidence-strength rule—trust the measurements when they deviate from the population norm, defer to video otherwise—which raises 0.75 to 0.79 (0.82 with the grounded model) where a nested global weight gives 0.75, and passes permutation, region-specificity, backbone and leave-one-out controls. The deviation signal is carried by head posture, uninformative alone but decisive given trunk inclination.

Keywords: Clinical priors · Gait analysis · Interpretability · Adult spinal deformity.

1 Introduction

Adult spinal deformity (ASD) alters walking in ways clinicians recognise at a glance from a lateral view: the trunk inclines forward and the pelvis, hips and knees compensate to keep the head over the feet [4,13]. Video-based assessment needs no markers, and it is natural to ask experts *where they look* and to feed that attention into a deep model as a grounding loss on its saliency or as a modality aligned to the video [9,11]—treating the annotation as something the network should *learn from*.

We report a controlled study on 79 patients (1,889 gait videos) annotated by two clinicians that reaches a different conclusion about what the annotation is for. (i) Used as supervision, it buys a verifiable attention map—our region-grounded 3D CNN aligns with clinician attention at 0.53 against 0.14 for Grad-CAM—but no accuracy: every end-to-end video model, from a Kinetics-pretrained 3D CNN to a video transformer, stays at 0.53–0.66 patient-level balanced accuracy, and four skeleton models are at chance. (ii) The annotation is a poor training signal: six distinct region combinations, 45.7% inter-clinician agreement. (iii) Yet it is highly informative about *what to measure*: five sagittal quantities at the marked regions, from 3D keypoints and a logistic regression, reach 0.72 (AUC 0.79), and no data-driven selection over 1,272 candidates beats them.

We therefore recast clinician attention from a training signal into a *measurement prior* (Fig. 1): the marked regions decide which quantities are measured, a readable linear model decides from them, and the annotation is never used at inference. The grounded video model and the five numbers err on different patients, and we combine them without a post-hoc weight: an *evidence-strength gate* measures how far a patient’s five quantities deviate from the population norm; unremarkable measurements carry no evidence and the video decides, otherwise the measurements decide. Learned inside the cross-validation, the rule lifts 0.75 to 0.79 (0.82 with the grounded model), and its signal is carried by head posture relative to the trunk—which is why, we suggest, clinicians mark the head.

Our contributions are: (1) a matched comparison of one annotation used as supervision versus as a measurement prior, under one protocol; (2) five clinician-defined measurements that beat all end-to-end models, with random-measurement, unmarked-region and candidate-bank controls showing that the value lies in the selection; (3) an evidence-strength gate validated by permutation, region-specificity, backbone and leave-one-out controls; (4) a full account of what did not work, in the supplement.

2 Related Work

Concept bottlenecks [9] and attention supervision make a network’s representation match expert concepts or regions; the payoff is a checkable explanation and the usual finding, as here, is unchanged accuracy. Post-hoc saliency such as Grad-CAM [14] is the common alternative; we measure that it does not recover clinician attention here. Kinetics-pretrained video networks [7,3,1] and skeleton graph networks [17,2] are the standard end-to-end choices, whereas clinical gait analysis relies on a few named kinematic quantities. Selecting measurements by ℓ_1 regression [15,5] and combining classifiers at the probability level [8] are classical; we test whether the expert boundary improves selection and gate the fusion by evidence strength rather than a learned mixture, in the spirit of interpretable-by-design models [12].

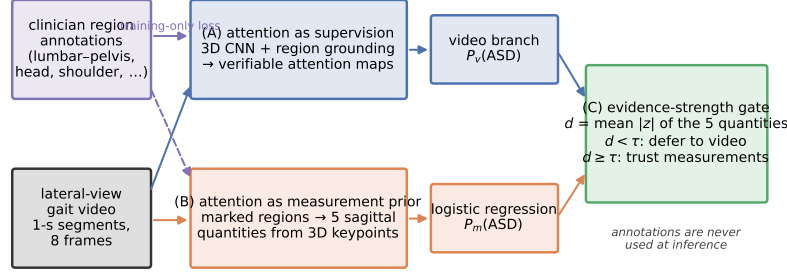


Fig. 1. Two uses of one clinician annotation. (A) As supervision: a region-grounded 3D CNN with verifiable attention (Sec. 3.1). (B) As a measurement prior: the marked regions define five sagittal quantities from 3D keypoints, classified by logistic regression (Sec. 3.2). (C) An evidence-strength gate decides per patient whether the measurements carry enough evidence to decide (Sec. 3.3).

3 Method

Data and annotation. Lateral-view walking videos of 79 patients (54 ASD, 25 non-ASD; the latter pools dropped-head syndrome and lumbar stenosis / hip osteoarthritis), 1,889 videos cut into 1-s segments of 8 frames (14,048 segments). Two clinicians marked, per patient, the regions they attend to from five candidates; at least one clinician marked lumbar-pelvis for 72% of patients, head for 47%, shoulder for 23%, wrist for 3% and foot for 1%. Only six combinations occur and the clinicians agree on 45.7% of region decisions. 2D keypoints come from YOLOv8-pose [6]; camera-frame 3D keypoints from SAM-3D-Body [16] exist for 67 patients, the primary cohort.

3.1 Attention as supervision (contrast)

A Kinetics-pretrained slow_r50 token grid is queried by $R=5$ learned concept vectors, one per region, by cross-attention; each yields an attention map A_r and a pooled feature. A presence head predicts whether region r is marked and the classifier consumes the presence-weighted concept features with a global feature. Training minimises $\mathcal{L}_{\text{cls}} + \lambda_p \mathcal{L}_{\text{presence}} + \lambda_g \mathcal{L}_{\text{ground}}$: a soft cross-entropy against the two-annotator targets $\{0, \frac{1}{2}, 1\}$ and a KL divergence between A_r and a Gaussian rendering of the region on the skeleton, weighted by the soft presence. Inference uses video only. Alignment is $\sum_n A(n) \hat{M}(n)$, $\hat{M} = M / \max M$; a uniform map scores 0.068.

3.2 Attention as a measurement prior

The annotation statistics single out three regions, for which we take the textbook sagittal readings, computed per segment from its 8 frames: *trunk lean*, the angle

of the shoulder-midpoint–hip-midpoint line to the vertical (forward positive); *shoulder offset*, the anterior offset of the shoulder midpoint from the hip midpoint over trunk length; *head forward*, the angle of the ear-midpoint–shoulder-midpoint line to the vertical; *maximum hip flexion*, 180° minus the smallest shoulder–hip–knee angle; and *hip range*, the within-segment range of that angle. A patient is the mean of each quantity over segments; the five values are standardised and classified by a class-balanced logistic regression ($C=1$). Nothing is tuned on the outcome. To test whether the selection matters we also build a *bank* of 1,272 candidates (pairwise landmark inclinations, offsets and distances, and chain angles, as segment means, extremes and ranges), each labelled by the regions it touches, and classify it by segment-level ℓ_1 logistic regression (C by an inner split) with the candidates restricted to the clinician regions, to their complement, or unrestricted, and by random forests and gradient boosting on all candidates.

3.3 Evidence-strength gating

Let P_v and P_m be the patient-level ASD probabilities of the video and five-number models, and d the patient’s mean absolute z -score of the five quantities with respect to the training patients. The fused probability is $P = (1 - w)P_v + wP_m$ with $w = w_{lo}$ if $d < \tau$ and $w = w_{hi}$ otherwise, $\tau = Q_q(d_{train})$, where (q, w_{lo}, w_{hi}) are chosen on the training patients of each outer fold from a small grid ($q \in \{0.2, \dots, 0.5\}$, $w_{lo} \in \{0, 0.25, 0.5\}$, $w_{hi} \in \{0.75, 1\}$) by balanced accuracy; the test fold only applies the rule: unremarkable measurements carry no evidence, so the video decides. Under the same nested protocol we compare a fixed $w=0.5$, a nested global w , stacking (logistic regression on the two logits) and a sigmoid gate $w = \sigma(\theta^\top u)$ on six reliability features (video confidence and segment disagreement, number of segments, measurement confidence, fraction of segments without 3D, and d), fitted by class-balanced likelihood with an inner-selected ℓ_2 penalty.

4 Experiments

Protocol. All splits are patient-disjoint. Video and skeleton models use one nested five-fold split (train/validation/test $\approx 60/20/20$; validation selects the checkpoint), 50 epochs, three seeds and identical optimisation (Adam, 10^{-4} , weight decay 10^{-3} , bf16); only the video transformer, unstable at that rate (0.571), is reported at 2×10^{-5} . Measurement models have no training randomness, so we re-draw the five-fold partition ten times and report mean $\pm$ s.d. (the fixed split is 0.03–0.05 optimistic for some variants). Fusion is bound to the fixed split and reported over three video branches (3D CNN, grounded, grounded without the grounding loss) $\times$ three seeds = nine groups. Twelve patients have no 3D keypoints and would be median-filled, so primary numbers use the 67 complete patients (46 ASD / 21 non-ASD); the 79-patient version is in the supplement. The metric is patient-level balanced accuracy (segment probabilities averaged

Table 1. Patient-level balanced accuracy and AUC, 67 patients. Video/skeleton rows: fixed split, 50 epochs $\times$ 3 seeds. Measurement rows: 10 random partitions. Fusion: fixed split $\times$ 3 seeds $\times$ 3 video branches.

Method	Input	Balanced acc.	AUC
3D CNN (slow_r50, K400)	video	0.644 $\pm$ 0.045	0.670 $\pm$ 0.021
TimeSformer-B (K400, 8 frames)	video	0.637 $\pm$ 0.055	0.716 $\pm$ 0.020
CLIP-style alignment (annotation at test)	video+annot.	0.589 $\pm$ 0.030	0.600 $\pm$ 0.027
3D CNN + region grounding (Sec. 3.1)	video	0.663 $\pm$ 0.030	0.694 $\pm$ 0.091
ST-GCN, 2D skeleton	skeleton	0.513 $\pm$ 0.010	0.563 $\pm$ 0.019
ST-GCN, 3D skeleton	skeleton	0.538 $\pm$ 0.026	0.557 $\pm$ 0.038
CTR-GCN, 2D skeleton	skeleton	0.491 $\pm$ 0.029	0.521 $\pm$ 0.066
CTR-GCN, 3D skeleton	skeleton	0.528 $\pm$ 0.065	0.553 $\pm$ 0.078
Random forest, 1,272 measurements	3D meas.	0.732 $\pm$ 0.019	0.736 $\pm$ 0.019
Gradient boosting, 1,272 measurements	3D meas.	0.738 $\pm$ 0.022	0.770 $\pm$ 0.018
L1 logistic regression, 1,272 measurements	3D meas.	0.730 $\pm$ 0.023	0.760 $\pm$ 0.019
L1 logistic regression, 300 within clinician regions	3D meas.	0.742 $\pm$ 0.032	0.755 $\pm$ 0.022
4 quantities at unmarked regions + LR	3D meas.	0.660 $\pm$ 0.037	0.679 $\pm$ 0.033
5 clinician-defined quantities + LR	3D meas.	0.717$\pm$0.017	0.787$\pm$0.016
same 5 quantities from the 2D skeleton	2D meas.	0.739 $\pm$ 0.031	0.785 $\pm$ 0.017
5 quantities + video, fixed $w=0.5$ / nested global w / sigmoid gate	fusion	0.717 / 0.745 / 0.773	
5 quantities + video, evidence gate	fusion	0.794 (grounded video: 0.819)	

per patient; chance 0.5) with AUC, bootstrap intervals and exact McNemar tests [10].

4.1 Main result

Every end-to-end model that consumes the video lands between 0.53 and 0.66 (Table 1): the 3D CNN at 0.644, the TimeSformer at 0.637 (best video AUC, 0.72), the grounded model at 0.663; a 2D CNN and a CNN-LSTM are lower (0.57, 0.53; supplement). The four skeleton models, two architectures $\times$ two skeleton sources, are at chance (0.49–0.54). The five clinician-defined quantities reach 0.717 ± 0.017 with AUC 0.787 ± 0.016 , the highest AUC in the table; from the 2D skeleton the same five give 0.739 ± 0.031 , so the result does not hinge on the 3D estimator. Four plausible quantities at the unmarked regions (knee flexion, step length, gait speed, arm swing) give 0.660, and adding them to the five *lowers* accuracy to 0.687. The same 3D joints that are at chance through a graph network are at 0.72 as five numbers: the signal is in what is measured, not in representation learning.

4.2 Is it the selection?

Fig. 2 asks whether *any* five measurements would do. Five drawn at random average 0.63 (all regions), 0.64 (clinician regions) and 0.60 (unmarked regions); the clinician-defined five sit at the 98th, 99th and 100th percentile of the 100 draws. Data-driven selection over all 1,272 candidates reaches 0.730 ± 0.023 (ℓ_1), 0.732 ± 0.019 (random forest) and 0.738 ± 0.022 (gradient boosting); restricting the candidates to the clinician regions gives 0.742 ± 0.032 and to their complement 0.710 ± 0.043 ($+0.03 \pm 0.05$, positive in 7/10 partitions): at the bank level the expert boundary does not improve selection beyond noise. The honest reading is parsimony—five named numbers equal the best of 1,272—and the annotation’s value is in naming them.

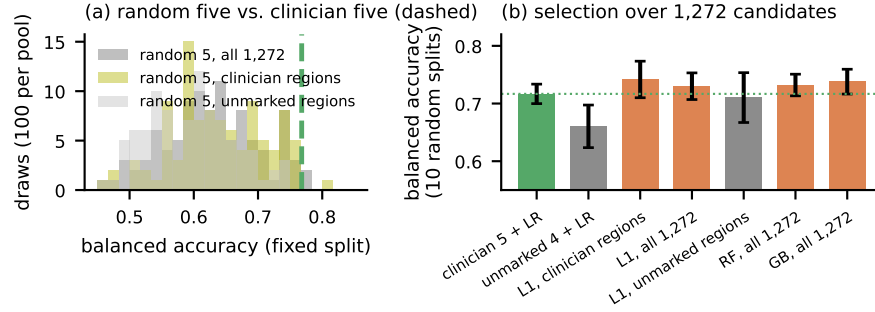


Fig. 2. (a) Five measurements drawn at random from the bank (100 draws per pool; fixed split) against the clinician-defined five. (b) Ten-partition mean $\pm$ s.d. of the five against data-driven selection over the 1,272 candidates.

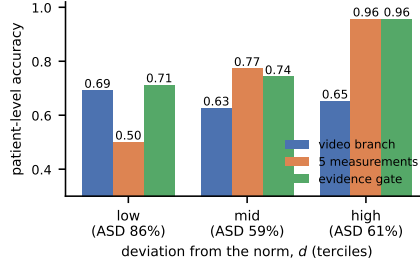


Fig. 3. Why the gate works: accuracy by tercile of d , pooled over the 9 fusion groups. Where measurements are unremarkable (left; 86% ASD) the five-number model is at chance and the video model is not; where they deviate the five numbers are almost always right.

4.3 Evidence-gated fusion and ablations

The video branch and the five numbers err on different patients: either being right would give 0.90. Table 2 (bottom) shows what is recoverable honestly. A fixed $w=0.5$ (0.717) and a nested global weight (0.745) do not beat the five numbers alone (0.751); stacking (0.731) and the sigmoid gate (0.773), whose coefficients collapse to a near-constant weight, do little better. The evidence-strength gate reaches 0.794 (grounded branch: 0.819), non-negative against the five numbers in all nine groups and against the nested weight in eight, and significant against the video branch in seven (McNemar $p \leq 0.05$). The rule is stable: τ at the 30–40th percentile of d , $w_{lo} \in \{0, 0.5\}$, $w_{hi} = 1$; any fixed $q \in [0.1, 0.4]$ gives 0.795–0.809.

Fig. 3 shows the mechanism. In the tercile of least deviation 86% of the patients are ASD but the five-number model scores 0.50: with normal-looking measurements it defaults to non-ASD, and these are ASD patients without visible sagittal compensation, whom the video model recognises at 0.69; in the other terciles the five numbers are right 0.77 and 0.96 of the time. Four controls close

Table 2. Ablations. Top: leave-one-out, region-group and single-quantity models (10 partitions). Bottom: fusion with the video branch under the same nested protocol, 9 groups.

Measurement ablation (10 splits)	Balanced acc.	AUC
all five	0.717 $\pm$ 0.017	0.787 $\pm$ 0.016
– trunk lean	0.682 $\pm$ 0.017	0.767 $\pm$ 0.016
– shoulder offset	0.696 $\pm$ 0.023	0.783 $\pm$ 0.014
– head forward	0.688 $\pm$ 0.031	0.739 $\pm$ 0.017
– max hip flexion	0.723 $\pm$ 0.020	0.790 $\pm$ 0.015
– hip range	0.731 $\pm$ 0.010	0.800 $\pm$ 0.010
– lumbar–pelvis group (3)	0.713 $\pm$ 0.019	0.793 $\pm$ 0.012
trunk lean alone	0.707 $\pm$ 0.011	0.778 $\pm$ 0.009
head forward alone	0.409 $\pm$ 0.070	0.399 $\pm$ 0.069
Fusion (3 video branches $\times$ 3 seeds)	Balanced acc.	
video branch alone	0.628	
5 quantities alone	0.751	
fixed $w=0.5$ / nested global w / stacking	0.717 / 0.745 / 0.731	
sigmoid gate (6 features)	0.773	
evidence gate	0.794	
evidence gate, deviation permuted (10 $\times$)	0.743	
evidence gate with TimeSformer video branch	0.785	

the argument: permuting d across patients drops the gate to 0.743 (10 permutations per group; the true value exceeds all 30 permutations in a 79-patient run); computing d from the four unmarked-region quantities gives 0.733, the permutation level; replacing the video branch by the TimeSformer gives 0.785; and leaving out head forward removes the gain entirely (0.794 $\rightarrow$ 0.729, five-number model at 0.740), whereas leaving out any other quantity changes the gate by at most 0.03.

The measurement ablation (Table 2, top) explains the last point. *Head forward alone* is uninformative (0.41; its direction even reverses because dropped-head patients in the non-ASD group lean the head further), yet removing it costs the most (0.717 $\rightarrow$ 0.688, AUC 0.79 $\rightarrow$ 0.74). Head posture is a *conditional* sign: given the trunk inclination, the head’s position relative to the trunk separates the forward-inclined ASD patient from the dropped-head non-ASD patient. Trunk lean alone gives 0.707, and dropping the whole lumbar–pelvis group costs almost nothing (0.713) because its three quantities are largely redundant: five numbers are, in effect, one strong sign plus one conditional sign.

4.4 Interpretability and negative results

The grounded branch keeps what the supervision bought: attention alignment 0.533 ± 0.001 (three seeds) against 0.135 ± 0.034 for Grad-CAM on the plain 3D CNN, 0.144 for a centre prior and 0.068 for a uniform map; 0.104 ± 0.012 without

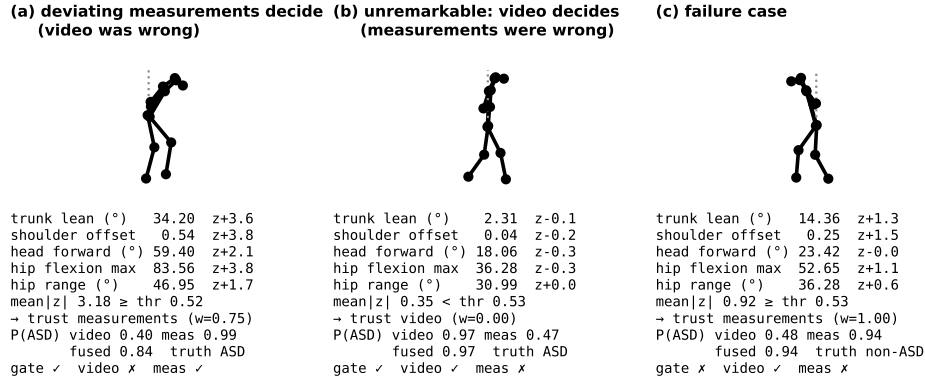


Fig. 4. Readable decisions for three patients (grounded video branch, one seed): the five quantities with z-scores, d against the fold’s threshold, the weight, and the branch and fused probabilities.

the grounding loss and 0.065 when grounded to random locations; region presence AP 0.81. The measurement branch needs no such machinery: every decision is five numbers, their deviation, and which branch was trusted (Fig. 4)—the classifier’s input, not a post-hoc rendering. Three negative lines bound the claims (supplement): feeding the five quantities *into* the video encoder, as auxiliary targets or by concatenation, never beats the video baseline in 255 runs except one variant at 0.715 ± 0.008 , still below the five numbers alone; a soft region-weighted ℓ_1 penalty on the bank is worse than a uniform one; and five uses of a vision-language model (frozen features, zero-shot diagnosis at AUC 0.44, prompting, attribute questions, explanations) are at or below the 3D CNN. The gate helps linear measurement branches but not the random forest or gradient boosting.

5 Discussion, Limitations and Conclusion

The same annotation helps as a prior and not as a signal because it is coarse as a label but precise as a pointer: “look at the trunk and the head” translates into two well-defined angles a linear model exploits, whereas a network given a saliency target learns where to look without learning what the look means. That head posture is a conditional sign is, we think, why clinicians mark the head; we flag it for clinical confirmation. The cohort is small (67 primary patients, 21 non-ASD): intervals are about ± 0.10 wide and single-seed differences below 0.08 are noise, so every number here is a three-seed or ten-partition mean. Fusion is evaluated on the fixed split only; the gate needs a linear, moderately accurate measurement branch; monocular 3D gives no coronal-plane quantities; data come from one centre.

In conclusion, clinician attention is most useful not as something a gait network should learn but as a specification of what to measure, and the evidence strength of those measurements tells us when to trust them over the video.

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